

Q-DECEM Revised Framework

Quarterly raw data



A1: raw predictors | Feature engineering -> A2



High-frequency grid data -> quarterly aggregation -> A3 | Engineering + grid fusion -> A4



Feature selection (FS1-FS5)



Five forecasting models (Ridge/RF/LightGBM/CatBoost/LSTM)



Nested expanding-window evaluation (outer + inner CV)



Pareto filtering



VIKOR (when >1 non-dominated)



Forecasting and interpretation